06E — Deep-Dive Feature Audit: Pills & Alchemy (Part 5 of 6)

Multi-Phase Specialty Elixirs (`PILL_V234_001..048`), Dynamic Elemental Fusion Pills (`PILL_V235_FUSION_*`) & Experimental Alchemy (`content.js`)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Master alchemists should physically tend two-phase elemental cauldrons using real element-matched herbs, and disciples who consume two-phase elixirs must genuinely experience Phase 1 (`Immediate Surge`) followed days later by Phase 2 (`Crystallization` or `Delayed Recoil`) regardless of how the pill was acquired!
- **Immutable Project Rules**:
    1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
    2. **Preserve exact 40 `<script>` tags** and load order in `index.html` / `test/load.js`.
    3. **Strict Closed-Loop Spirit-Stone Conservation** (`Debit === Credit` with `W.world.resourcePool.stones`).

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1. Architectural Scope of Group 5 (`06E`)

Group 5 audits the advanced multi-stage and experimental alchemy systems introduced across `v1.8` (`content.js`), `v2.0.34–v2.0.35`, and `v3.5.0` (`PROP-29`):

1. **48 Authored Multi-Phase Specialty Elixirs (`PILL_V234_001..048`)** & `EFX_V234_MULTI_PHASE_ELIXIR` / `EFX_V234_MULTI_PHASE_PILL`.
2. **Dynamic Elemental-Reaction Fusion Elixirs (`PILL_V235_FUSION_<ELEMENTS>`)**: `ECO.brewMultiphaseElementalPill` & `ECO.brewMultiPhaseSpecialtyPill` (`js/economy.js:1170–1242`) persisted via `W.sect.customPills`.
3. **Delayed Phase-2 Runtime Ticker**: `CULT.tickMultiPhasePills` (`js/cultivation.js`, invoked by `SIM.hourly`) vs `ECO.useConsumableItem` (`economy.js:1357–1381`) vs `ECO.applyPill` (`economy.js:354–364`).
4. **Experimental Alchemy & Formula Discovery (`js/content.js`)**: `experimentId` research, secret `RECIPE_SCROLL` hoarding, peer scroll sales, and sect recipe donation.

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2. Critical Defect 1: `[PILL-MP-01]` Split-Brain Dead Queue (`pendingPillPhases` vs `multiPhasePills`) — Phase 2 NEVER Fires (`0%`) When NPCs Consume Multi-Phase Pills Autonomously!

2.1 Exact Code Proof (`js/economy.js:354–364` vs `1367–1381` & `v350_macro_jianghu_and_alchemy_suite.js:154`)

Compare the **two separate functions** that queue Phase 2 of a multi-phase pill:

1. **`ECO.useConsumableItem` (`js/economy.js:1367–1381` — used ONLY by UI button / `v350` unit test!)**:

```
JavaScript
npc.ext.multiPhasePills = Array.isArray(npc.ext.multiPhasePills) ?
npc.ext.multiPhasePills : [];
npc.ext.multiPhasePills.push({
  itemId: def.id, nameTh: def.nameTh, startedDay: dayNow, startedTick: tNow,
  triggerDay: dayNow + delayDays, triggerTick: tNow + delayTicks, phase2: p2,
resolved: false
});
```

   `CULT.tickMultiPhasePills` (called in `SIM.hourly`) reads **`npc.ext.multiPhasePills`**!

2. **`ECO.applyPill` (`js/economy.js:354–364` — used by ALL autonomous NPC pathways: `CPE.autoUse`, `A.personalPillUse`, `CPE.fightUse`, `BN.prepare`, `ECO.usePill`!)**:

```
JavaScript
if (Array.isArray(e.phases) && e.phases.length > 1) {
  npc.ext = npc.ext || {};
  npc.ext.pendingPillPhases = Array.isArray(npc.ext.pendingPillPhases) ?
npc.ext.pendingPillPhases : [];
  npc.ext.pendingPillPhases.push({ pillKey: key, triggerDay: ..., ... });
}
```

- **ZERO functions anywhere in the entire repository read `npc.ext.pendingPillPhases`!**
- In live gameplay, whenever any NPC buys a `PILL_V234_*` or `PILL_V235_FUSION_*` from the Sect Shop and drinks it via `CPE` / `ECO.usePill`, **Phase 2 is dumped into the dead `pendingPillPhases` array, never triggers (`0%`), and leaks unbounded memory into the save file!**

2.2 Exact Fix

Delete `pendingPillPhases` completely (migrating any legacy entries on load) and push standardized `{ itemId, nameTh, startedDay, startedTick, triggerDay, triggerTick, phase2, resolved: false }` entries directly into **`npc.ext.multiPhasePills`** inside `ECO.applyPill` (with a guard `if (!ctx || !ctx._fromUseConsumable)` so it never double-queues)!

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3. Critical Defect 2: `[PILL-MP-02]` Double Phase-1 Payout in `ECO.useConsumableItem` (`economy.js:1192–1198, 1344–1366`) & Swapped `ST.addStatus` Duration Argument in `tickMultiPhasePills`!

3.1 Bug A — Phase 1 Applied Twice in the Same Millisecond (`economy.js:1344–1366`)

In `ECO.brewMultiphaseElementalPill` (`lines 1192–1198`) and authored `PILL_V234_*` items:

- Top-level fields define `healPct: 0.42, qiPct: 0.45, cultExp: 210, grantBuff: 'alchemicalHarmony'`.
- `phases[0]` *also* defines `{ phase: 1, hpHealPct: 0.40, qiRestorePct: 0.42, cultExpAdd: 95, grantBuff: 'elementalOvercharge' }`.
- In `ECO.useConsumableItem` (`lines 1345–1366`), lines `1345–1354` apply the top-level `healPct`/`qiPct`/`cultExp`/`grantBuff`, **and then lines `1358–1366` immediately apply `phases[0]` on top of them**—granting **`+82%` HP, `+87%` Qi, `+305` CultExp** in a single swallow and immediately overwriting `def.grantBuff` with `p1.grantBuff`!
- **Fix**: When `Array.isArray(def.phases) && def.phases.length >= 2`, treat `def.phases[0]` as the authoritative Phase-1 payload (skipping duplicate top-level `healPct`/`qiPct`/`cultExp` when `p1` already specifies them)!

3.2 Bug B — Swapped Potency vs Duration in `CULT.tickMultiPhasePills` (`js/cultivation.js`)

- `ST.addStatus(u, k, dur, extra, now)` (`stats.js:241`) takes **`dur` (integer ticks) as Argument 3** and `extra` as Argument 4.
- When `tickMultiPhasePills` applies a Phase-2 recoil debuff (`e.g. weaken` / `slow`), passing decimal potency (`e.g. 0.12`) in Argument 3 causes `clamp(Math.round(0.12), 1, 2880)` to round duration down to **`1` tick (1 minute)** instead of the intended `p2.debuffDur || 120` ticks (`{ potency: 0.12 }` in Argument 4)!

3.3 Bug C — `'DARK_YIN'` Hardcode Breaks Yin-Yang Zero-Toxicity Alchemy (`economy.js:1184`)

- Line 1184 checks `(els.includes('LIGHT_YANG') && els.includes('DARK_YIN'))`, while all real game herbs and subpaths use **`'YIN_SHADOW'`** (`03B`)! Accept `(els.includes('DARK_YIN') || els.includes('YIN_SHADOW'))` and normalize `'YIN_SHADOW'` inside `resolveElementalReactionSpec` so Taiji Zero-Toxicity Elixirs work with real game herbs!

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4. Critical Defect 3: `[PILL-MP-03]` Tier-7 Mythic Bypass, Priceless Herb Destruction, Stone-Pool Leak & `0%` Autonomous Brewing in `brewMultiPhaseSpecialtyPill` (`economy.js:1160–1242`)!

Inspecting `js/economy.js:1160–1242` reveals **four severe economic & simulation flaws**:

1. **Instant 0-Minute Tier-7 Mythic Pill Conjuration Using `2` Tier-1 Weeds (`economy.js:1213–1228`)!**:
   - `ECO.brewMultiPhaseSpecialtyPill(W, alch, 'PILL_V234_048')` brews **any** of the 48 specialty pills—including **Tier-6/7 Mythic Elixirs (`value = 800+` stones)**—for a flat **`2` arbitrary Tier-1 Dew-Grass weeds + `25` Spirit Stones**, in **`0` minutes**, with **`100%` guaranteed success**, and **zero check on `alchemist.h.skills.ALCHEMY`, `W.sect.recipeTier`, or `alchemy.level`**!
2. **Blind `consumeSectHerbs(W, herbNeed)` Eats Priceless Tier-5/7 Herbs & Ignores Declared Elements (`economy.js:1160–1168, 1180`)!**:

```javascript
for (const k of getHerbStockKeys(W)) {
  const takeAmt = Math.min(have, rem);
  if (takeAmt > 0 && ECO.take(W, k, takeAmt)) rem -= takeAmt;
}
```

    - `consumeSectHerbs` grabs the **first keys in insertion order** without sorting by `tier ASC`, without checking `ECO.reserveFloor(W, k)`, and without matching `herbElements`! If `MAT_HERB_SNOW_LOTUS_5` or `MAT_HERB_CHAOS_ROOT_7` happens to be first in `W.sect.stock`, brewing a 20-stone pill **destroys a 1,500-stone Tier-7 Chaos Root**!
3. **Closed-Loop Spirit-Stone Leak (`economy.js:1181` vs `1226`)**:
    - `brewMultiPhaseSpecialtyPill` (`line 1226`) returns `stoneCost` to `W.world.resourcePool.stones`, **but `brewMultiphaseElementalPill` (`line 1181`) calls `ECO.pay(W, 20)` without crediting `W.world.resourcePool.stones`**, permanently deleting 20 stones per brew from the closed-loop world!
    - Also, line `1229` (`W.sect.customPills[targetDef.id] = targetDef`) needlessly copies static `STATIC_ITEM_IDS` (`PILL_V234_*`) into `W.sect.customPills`, bloating save files!
4. **`0%` Autonomous Alchemist Usage (`UI-Button Only!`)**:
    - Neither `brewMultiPhaseSpecialtyPill` nor `brewMultiphaseElementalPill` is ever called by NPC AI!
    - **Emergent Fix**: Allow `ELIXIR_DAO` / `NINE_REVOLUTIONS_ALCHEMIST` / `alchemy` Hall Elders (`ALCHEMY >= 45`, `alchemy.level >= 2`) during `A.brew` to genuinely brew tier-eligible (`tier <= recipeTier`, requiring real tier-matched elemental herbs above `reserveFloor` and real cauldron hours!) Two-Phase & Elemental Fusion Elixirs into `W.sect.stock`—and teach `CPE.pickPurchase` to value `e.phases` pills for disciples preparing for Closed-Door Seclusion or breakthroughs!

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5. Logic & Context Flaw: `[PILL-MP-04]` Experimental Alchemy (`js/content.js`) Runs as a Midnight Telepathic Script Instead of Physical Cauldron Research!

- In `js/content.js` (`C.daily`), discovering experimental pill recipes (`experimentId`) runs instantaneously at midnight via `W.npcs.find(...)` (biasing `W.npcs[0]`) without walking to `alchemy` or spending `energy`, while disciples hoarding secret

`RECIPE_SCROLL`s sell them telepathically (`02A`) instead of brewing scarce high-margin specialty pills via `A.personalBrew` to sell at `market`!

- Convert alchemy experimentation into a physical `AI.ACTS.researchAlchemy` cauldron activity driven by `curiosity`, `inventiveness`, `ALCHEMY` skill, and `ELIXIR_DAO` / `CRAFT_CREATION` aspirations!

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6. Mandatory Verification Suite Additions (`06E`)

1. **Autonomous `ECO.usePill` $\to$ Phase-2 Trigger Test**: Have an NPC consume `PILL_V234_001` via **`ECO.usePill` / `CPE.autoUse`** (not only `useConsumableItem`!), advance `W.t` past `triggerTick`, run `SIM.hourly(W)`, and assert Phase 2 fires and `pendingPillPhases` does not exist.
2. **Single Phase-1 Payout Test**: Consume a `PILL_V235_FUSION_*` pill via `useConsumableItem` and assert Phase-1 `cultExp` / heal is applied **once**, not doubled.
3. **Herb Protection & Closed-Loop Conservation Test**: Put `MAT_HERB_CHAOS_ROOT_7` (`qty: 1`) and `MAT_HERB_DEW_GRASS_1` (`qty: 20`) in `W.sect.stock`, brew an elemental pill, and assert `CHAOS_ROOT_7` is untouched and `W.sect.stones + W.world.resourcePool.stones` remains invariant!